

$$\bar{h} = \frac{1}{x} \int_0^x h dx$$

$$\bar{h} = \frac{1}{x} \int_0^x \frac{Ck}{U^{1/2}} x^{(m-1)/2} dx = \frac{2}{m+1} \frac{Ck}{U^{1/2}} x^{(m-1)/2}$$

$$\boxed{\frac{\bar{h}}{h} = \frac{2}{m+1}} \Rightarrow h \text{ is the local heat transfer coefficient } (h(x))$$

Note in real experiments, for $Pr \approx 1$ ($0.5 < Pr < 10$) and $m=1$:

$$\boxed{Nu_x = 0.57 Re_x^{1/2} Pr^{0.4}} \Rightarrow \text{Jet impinging on a wall (2-D)}$$

note: $Re_x = \frac{U_\infty x}{\nu}$

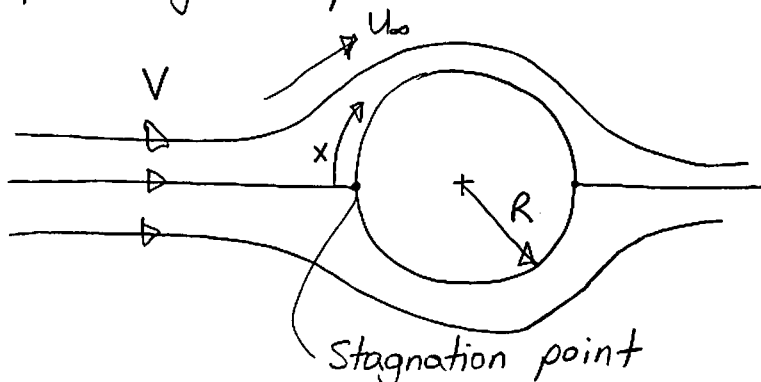
For a 3-D stagnation point:

$$\boxed{Nu_x = 0.76 Re_x^{1/2} Pr^{0.4}} \Rightarrow \text{Jet impinging on a wall (3-D)}$$

Makes sense that $h_{x,3D} > h_{x,2D}$ since we create a third dimension to dissipate energy & momentum. This is similar to turbulence!

Cylinders and Spheres

To use the above jet impingement solutions ($Nu_x Re_x^{-1/2} = \text{const}$) it is useful to know the variation of U_∞ near the stagnation point: Most solutions can be approximated by a cylinder or sphere geometry.



$$U_\infty = \frac{2Vx}{R} \text{ for small } \frac{x}{R}$$

$\hookrightarrow$ For a cylinder

We know our solution for a 2-D stagnation point is: ($m=1$)

$$Nu_x = 0.57 Re_x^{1/2} Pr^{0.4}$$

$$Re_x = \frac{U_\infty x}{\nu}, \quad Nu_x = \frac{hx}{k} \Rightarrow \text{Convert to } R \text{ length scale}$$

$$\frac{hx}{k} = 0.57 \left(\frac{U_\infty x}{\nu} \right)^{1/2} Pr^{0.4}$$

$$\text{For a cylinder, } U_\infty = \frac{2Vx}{R} \quad (V = \text{far field free stream velocity})$$

$$\frac{hx}{k} = 0.57 \left(\frac{2Vx^2}{\nu R} \right)^{1/2} Pr^{0.4}$$

Multiply both sides by R

$$\frac{hR}{k} = 0.57 (2)^{1/2} \left(\frac{\nu R}{V} \right)^{1/2} Pr^{0.4}$$

$$\boxed{Nu_R = \frac{hR}{k} = 0.81 Re_R^{1/2} Pr^{0.4}} \quad 0.5 < Pr < 10, \quad 0 < Re_R < 5 \times 10^5 \text{ (Laminar)}$$

$\Rightarrow$ For a cylinder near stagnation pt.

Note, we can do the same analysis for a sphere:

$$U_\infty = \frac{3Vx}{2R} \quad \text{for small } \frac{x}{R} \Rightarrow \text{Sphere (3-D)}$$

Following the same steps as above, we obtain:

$$\boxed{Nu_R = 0.93 Re_R^{1/2} Pr^{0.4}} \quad 0.5 < Pr < 10, \quad 0 < Re_R < 5 \times 10^5 \text{ (Laminar)}$$

$\rightarrow$ For a sphere near stagnation point.

Note, these are not equivalent to $\overline{Nu}_R = \frac{hR}{k}$!

Note both of these solutions are for $Re_R < 5 \times 10^5$ which is well beyond when vortices form ($Re_R \approx 40$). However, vortices are on the trailing edge, and these apply to the leading edge.

Blowing & Suction at the Wall (Similarity Solutions)

So far, we have always assumed that $v|_{y=0} = 0$ since our wall was impermeable.

A number of applications use suction ($v|_{y=0} < 0$) or blowing ($v|_{y=0} > 0$). For example porous surfaces or surfaces with condensation or evaporation.

Our fundamental equations don't change, mainly:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \Rightarrow \text{Momentum}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} \Rightarrow \text{Energy}$$

$\Updownarrow$ Using our streamfunction formulation (9)

$$f''' + \frac{1}{2}(m+1)ff'' + m(1-(f')^2) = 0 \Rightarrow \text{Momentum} \quad (1)$$

$$\theta'' + \frac{1}{2}Pr(m+1)f\theta' = 0 \Rightarrow \text{Energy} \quad (2)$$

Note, our boundary conditions before were: (in momentum)

Dimensional

$$1) u(y=0) = 0$$

$$2) u(y \rightarrow \infty) = U_\infty$$

$$3) v(y=0) = 0$$

Similarity Solution ($f' = \frac{u}{U_\infty}$)

$$1) f'(\eta=0) = 0 ; \quad \eta = y \sqrt{\frac{U_\infty}{\nu x}}$$

$$2) f'(\eta \rightarrow \infty) = 1$$

$$3) v = \frac{1}{2} \left(\frac{\nu U_\infty}{x} \right)^{1/2} (\eta f' - f) \quad [\text{From mass conserved.}]$$

$$\text{at } y=0, \eta=0 \text{ so } f(0)=0 \quad \text{Pg. 48}$$

However, now that we have arbitrary blowing or suction, our third boundary condition must change if we want to attempt a similarity solution.

Note, for an arbitrarily prescribed $v|_{y=0}$, equations ① and ② on the previous page must be solved numerically.

For our case, we seek a similarity solution knowing:

$$v = -\frac{2\psi}{2x}, \quad \psi = \sqrt{U \times U_\infty} \cdot f, \quad u_\infty = Cx^m$$

If we combine these, we obtain

$$v = -f \frac{m+1}{2} x^{(m-1)/2} \sqrt{Cv} - C_y \frac{m-1}{2} f' x^{m-1}$$

But we know for our case, at $y=0$ or $\eta=0$, $V=V_s$

$$v|_{\eta=0} = V_s = -f \frac{m+1}{2} x^{(m-1)/2} \sqrt{Cv} - \underbrace{C(0) \frac{m-1}{2} f'(0) x^{m-1}}_0$$

We know $C = u_\infty / x^m$

$$V_s = -\frac{m+1}{2} f(0) \frac{u_\infty}{\sqrt{u_\infty x / U}}$$

or

$$f(0) = -\frac{2}{m+1} \frac{V_s}{u_\infty} \sqrt{\frac{U_\infty x}{U}} = -\frac{2}{m+1} \frac{V_s}{u_\infty} Re_x^{1/2} \Rightarrow \text{Our new boundary condition.}$$

Since we know that f is a similarity solution, and f is a function of η only, for $\eta=0$ ($y=0$), f must be a constant for all solutions.

So to obtain a true similarity solution, our b.c.'s become:

$$f(0) = -\frac{2}{m+1} \frac{V_s}{U_\infty} Re_x^{1/2} = \text{constant} \quad (3)$$

$$f'(0) = 0 \quad (4)$$

$$f(\eta \rightarrow \infty) = 1 \quad (5)$$

Equation (3) implies that to obtain a similarity solution,

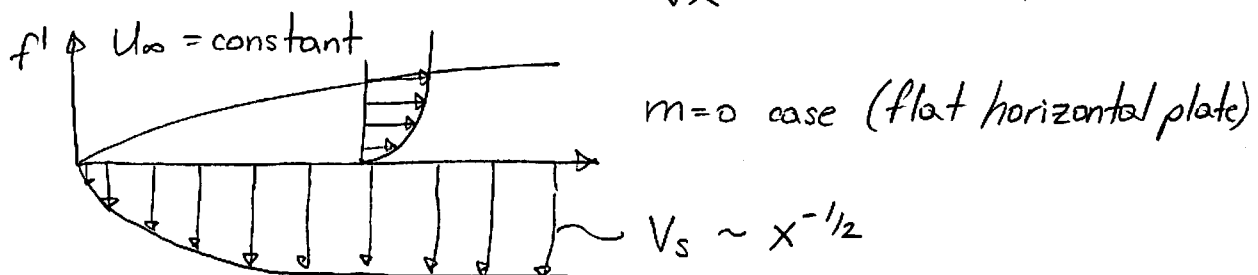
$$\begin{aligned} f(0) = \text{constant} &= -\frac{2}{m+1} \frac{V_s}{U_\infty} \sqrt{\frac{U_\infty x}{\nu}} \Rightarrow U_\infty = C x^m \\ &= -\frac{2}{m+1} \frac{V_s}{C x^m} \sqrt{\frac{C x^{m+1}}{\nu}} \\ &= -\frac{2}{m+1} \frac{C^{1/2} x^{1/2} x^{m/2}}{C^{1/2} x^{m/2}} \cdot \frac{V_s}{\sqrt{\nu}} = -\frac{2}{m+1} C^{-1/2} x^{-(m-1)/2} \frac{V_s}{\sqrt{\nu}} \end{aligned}$$

$$\text{constant} = -\frac{2}{m+1} x^{-(m-1)/2} \cdot \frac{V_s}{\sqrt{\nu}}$$

$$\boxed{V_s \sim x^{(m-1)/2}} \text{ for this equation to work for all } x$$

The other possible similarity solution is for $m=1$ (stagnation)

For $U_\infty = \text{constant}$, $m=0$, $V_s \sim \frac{1}{\sqrt{x}}$ for similarity solution:



Typically, similarity solutions are characterized by:

$$\boxed{\frac{V_s}{U_\infty} Re_x^{1/2}} \equiv \text{Suction or Blowing Parameter} \quad (73)$$

We use this parameter since we know in our solution that:

$$\text{Constant} = - \underbrace{\frac{2}{m+1}}_{\text{constant}} \cdot \underbrace{\frac{V_s}{U_\infty}}_{\text{constant}} Re_x^{1/2} \Rightarrow \text{from eq. (3)}$$

So for any m , and suction or blowing parameter, equation (1) can be numerically solved & results tabulated.

$\frac{V_s}{U_\infty} Re_x^{1/2}$	$f''(0) = \frac{1}{2} C_{f,x} Re_x^{1/2}$	($m=0$)
- 2.5	2.59	Suction
- 0.75	0.945	
- 0.25	0.523	
0	0.332	Impermeable wall
+ 0.25	0.165	Blowing or Injection
+ 0.375	0.094	
+ 0.5	0.036	
+ 0.619	0	Separation

It's worth noting the order of magnitude of V_s here:

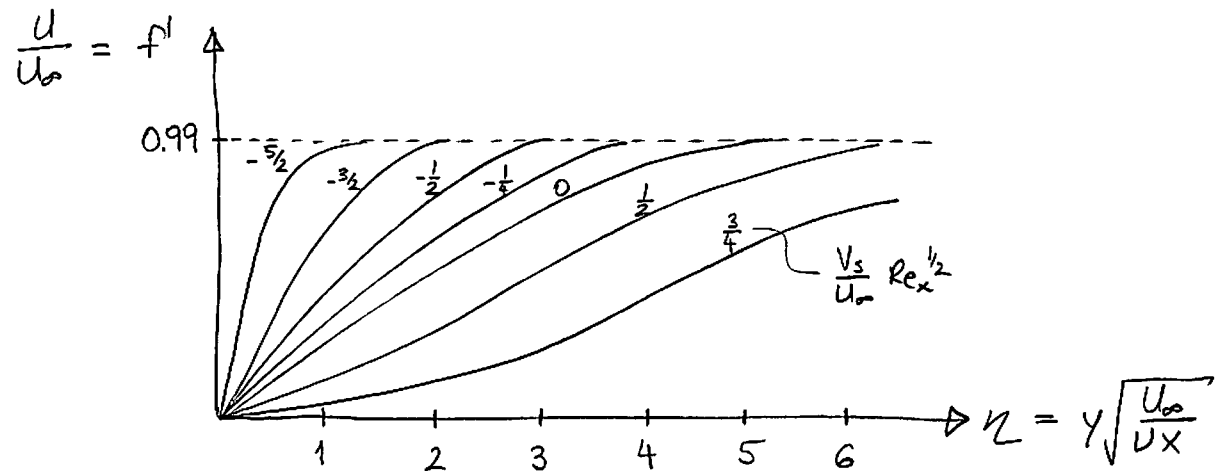
$$\frac{V_s}{U_\infty} Re_x^{1/2} \sim 1 \Rightarrow V_s \sim \frac{U_\infty}{Re_x}$$

Looking at v in our boundary layer (cons. of mass)

$$v \sim \frac{U_\infty}{Re_x^{1/2}} (2f' - f) \Rightarrow \text{We know } f' \& f \sim 1$$

So $V_s \sim v \Rightarrow$ So for the above suction velocity results, the suction parameters are such that the suction velocity, V_s is on the same order as the transverse velocity inside the b.l., v . Note, if $\frac{V_s}{U_\infty} Re_x^{1/2} \rightarrow 0$, $V_s \ll v$ and the suction solution is not very different from the flat plate solution.

If we plot our results,



From the solutions, we see that:

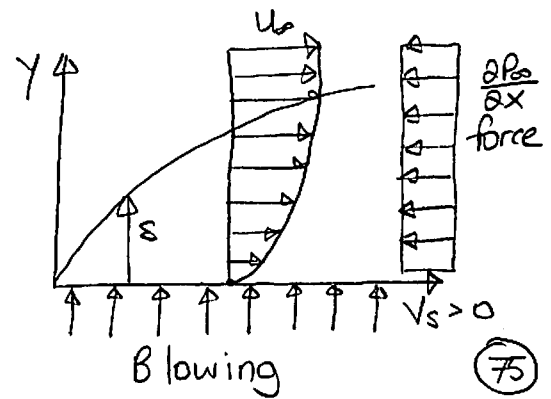
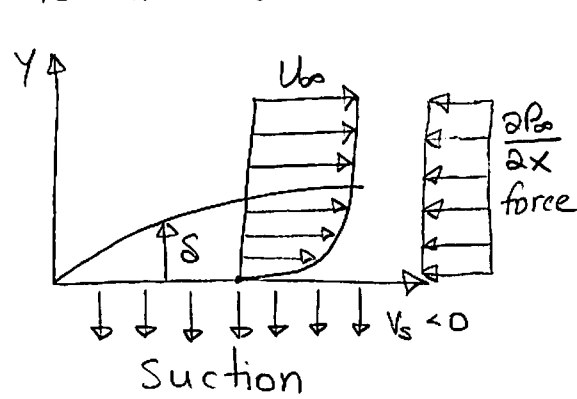
If $\frac{V_s}{U_\infty} Re_x^{1/2} < 0$ (Suction), δ decreases, τ increases

If $\frac{V_s}{U_\infty} Re_x^{1/2} > 0$ (Blowing), δ increases, τ decreases

At $\frac{V_s}{U_\infty} Re_x^{1/2} = 0.619 \Rightarrow$ Separation occurs $\Rightarrow \left. \frac{\partial u}{\partial y} \right|_{y=0} = 0$

Suction tends to increase the boundary layer attachment and delay separation. This comes at the cost of increased shear and drag. Sometimes used on aircraft wings: F-16XL
General Dynamics

Think of it as the suction pulls in the U_∞ stream closer to the wall, which can act against the $\frac{\partial p_0}{\partial x}$ adverse momentum close to the wall.



A simple particular solution can be obtained far from the leading edge.

Assuming $U_\infty = \text{constant}$, $m = 0$ (flat plate), $\frac{\partial p_\infty}{\partial x} = 0$
Our B.C.'s are:

$$\begin{aligned} u(y=0) &= 0 \quad (\text{no slip}) \\ v(y=0) &= -V_s \quad (\text{suction}) \\ u(y \rightarrow \infty) &= U_\infty \end{aligned}$$

One particular solution that satisfies these b.c.'s is where $u \neq f(x)$ or $u = f(y)$ only. Our momentum equation becomes:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = - \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (1)$$

From continuity: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial v}{\partial y} = 0 \Rightarrow v(y) = \text{constant}$

Since $v|_{y=0} = -V_s \Rightarrow v(y) = -V_s \quad (2)$

Substituting (2) into (1), we obtain:

$$-V_s \frac{\partial u}{\partial y} + \nu \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \text{Solvable ODE, second order, linear}$$

$$u(y) = A e^{-\lambda y} + B e^{+\lambda y} \quad \frac{\partial^2 u}{\partial y^2} + \frac{V_s}{\nu} \frac{\partial u}{\partial y} = 0$$

$$\lambda^2 + \frac{V_s}{\nu} \lambda = 0 \Rightarrow \lambda(\lambda + \frac{V_s}{\nu}) = 0 \Rightarrow \lambda = -\frac{V_s}{\nu} \text{ or } \lambda = 0$$

$$u(y) = A e^{+\frac{V_s}{\nu} y} + B e^{-\frac{V_s}{\nu} y} + C \Rightarrow \text{Since } u(y \rightarrow \infty) = U_\infty, A = 0$$

$$u(y \rightarrow \infty) = U_\infty = B e^{-\frac{V_s}{\nu} (\infty)} + C \Rightarrow C = U_\infty$$

$$u(y=0) = 0 = B e^{(0)} + U_\infty \Rightarrow B = -U_\infty$$

$$\therefore \boxed{u(y) = U_\infty (1 - e^{-\frac{V_s y}{\nu}})} \Rightarrow \text{Asymptotic solution far from the leading edge.}$$

Note, this solution is only valid for: $\boxed{\frac{V_s}{U_\infty} Re_x^{1/2} > 2}$

Our previous tabulated solutions for $m=0$ will fall on this curve for large x .

Heat Transfer (Suction & Blowing)

Our energy equation remains the same as before:

$$\Theta'' + \frac{1}{2} Pr(m+1)f\Theta' = 0 \quad ; \quad \Theta(\eta) = \frac{T - T_\infty}{T_w - T_\infty}$$

B.C.'s: $\Theta(\eta=0) = 0$
 $\Theta(\eta \rightarrow \infty) = 1$

To solve, we substitute our hydrodynamic solution, f , into the ODE above and numerically solve it with the same b.c.'s as before. Note, f is not unique and changes for each case of blowing & suction parameter, and m .

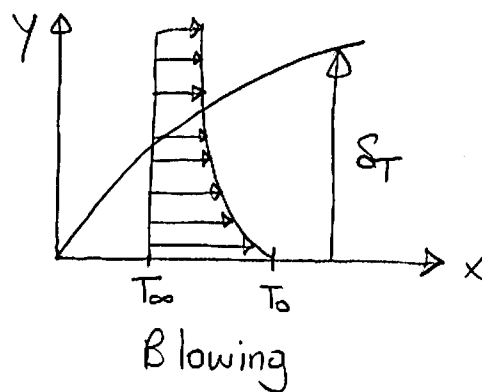
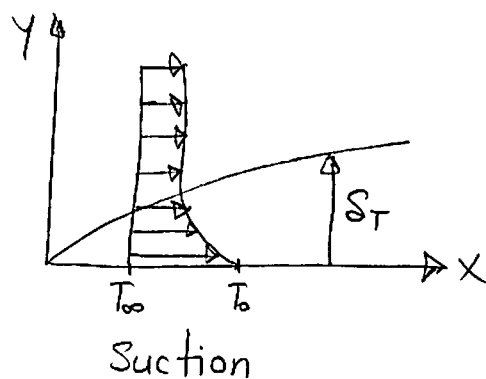
Looking at our case of $m=0$, $V_s \sim x^{-1/2}$

$\frac{V_s}{U_\infty} Re_x^{1/2}$	$Nu/Re_x^{1/2}$		(For $m=0$)
	$Pr = 0.7$	$Pr = 0.9$	
-2.5	1.85	2.59	Suction
-0.75	0.722	0.945	
-0.25	0.429	0.523	
0	0.292	0.332	Impermeable Wall
+0.25	0.166	0.165	Blowing
+0.375	0.107	0.0937	
+0.5	0.0517	0.0356	
+0.619	0	0	Separation

Note the $\frac{V_s}{U_\infty} Re_x^{1/2} = 0$ solution is the Blasius-Pohlhausen solution we've done before.

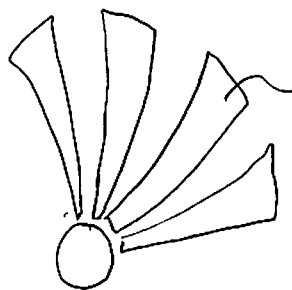
Note, $Nu_x = \frac{hx}{k} \uparrow$ for $\frac{V_s}{U_\infty} Re_x^{1/2} < 0$ (Suction, δ_T smaller)

$Nu_x = \frac{hx}{k} \downarrow$ for $\frac{V_s}{U_\infty} Re_x^{1/2} > 0$ (Blowing, δ_T larger)



Since $h \sim k \frac{\Delta T}{\delta_T} \Rightarrow h_{\text{suction}} > h_{\text{blowing}}$

Note, this can be a means of cooling. For example



Turbine blades on a jet or a generator
Blades are colder than the hot gases ($T_0 < T_\infty$). So blowing will decrease h and cool the blade. In this case, blowing is advantageous.

Also note, these solutions are valid for the same fluid being injected or sucked. If not $\Rightarrow$ mass transfer is important!

Same as previously derived: pg. 68 & 69 of notes

$$\boxed{\frac{\bar{h}}{h} = \frac{2}{m+1}}$$